

The unit-sphere chord metric determines the ambient normed space

MathlibAnnex.Sphere.nonempty_isometryEquiv_iff_nonempty_linearIsometryEquiv_of_finiteDimensional

theorem

Classifies existence of ambient linear isometries from the unit-sphere chord metric.

Statement

Let X, Y be real normed vector spaces and assume

$$\dim_{\mathbb{R}} X < \infty \quad \text{or} \quad \dim_{\mathbb{R}} Y < \infty.$$

Let $S_X = \{x : \|x\|_X = 1\}$ and $S_Y = \{y : \|y\|_Y = 1\}$, with chord distances $d_X(u, v) = \|u - v\|_X$ and $d_Y(u, v) = \|u - v\|_Y$. A sphere isometry equivalence is a bijection $\Delta : S_X \rightarrow S_Y$ preserving these distances. A real linear isometry equivalence is a real linear bijection $A : X \rightarrow Y$ satisfying $\|Ax\|_Y = \|x\|_X$ for every x . Then

$$(\exists \text{ a sphere isometry equivalence } S_X \rightarrow S_Y) \iff (\exists \text{ a real linear isometry equivalence } X \rightarrow Y).$$

Assumptions

The real normed-space structures and the finite-dimensional disjunction are the inputs. Neither positive dimension, completeness nor initial finite-dimensionality of both spaces is an extra assumption.

Conclusion

The theorem equates existence statements. It does not assert that the recovered ambient map restricts to a specified Δ . Zero-dimensional spaces and empty unit spheres are included.

Proof route

Transfer finite dimension in either direction, separate zero dimension, transport the positive-dimensional case to norm models, recover a linear isometry there, and compose back. Conversely restrict an ambient linear isometry.

▼ Proof steps

- 1. Transfer dimension in the applicable direction.** Given Δ , if X is finite-dimensional, finite dimension transfer along a sphere isometry makes Y finite-dimensional. If Y is finite-dimensional, apply it to Δ^{-1} to make X finite-dimensional. Then equality of dimensions gives $\dim_{\mathbb{R}} X = \dim_{\mathbb{R}} Y = n$. If $n = 0$, both spaces consist only of zero; the unique map $0 \mapsto 0$ is a real linear isometry equivalence. The sphere restriction is the empty bijection. the finite-dimensional construction including zero separates this case before using positive-dimensional coordinates. The dimension-transfer input is made explicit by the radial map

$$R_\Delta(x) = \begin{cases} 0, & x = 0, \\ \|x\|_X \Delta(x/\|x\|_X), & x \neq 0. \end{cases}$$

Its inverse is $R_{\Delta^{-1}}$. The radial estimates are $\|R_\Delta(x) - R_\Delta(y)\|_Y \leq 3\|x - y\|_X$ and the analogous inverse inequality. Both maps are continuous, and these Lipschitz bounds in both directions give equality of Hausdorff dimensions. A finite-dimensional X is locally compact; its radial homeomorphism makes Y locally compact, which forces finite-dimensionality for a real normed space. With both finite-dimensional, their Hausdorff dimensions equal their real dimensions. This use of the radial map proves dimension transfer; it does not identify the recovered linear A with R_Δ .

2. Use one coordinate system on each space. For $n = m + 1 > 0$, choose continuous real linear equivalences $e_X : \mathbb{R}^n \rightarrow X$ and $e_Y : \mathbb{R}^n \rightarrow Y$. Define

$$p(x) = \|e_X x\|_X, \quad q(x) = \|e_Y x\|_Y.$$

The operator bounds for these equivalences and their inverses give positive comparisons with the reference sup norm. Transport the given sphere map by

$$\Delta_M = (e_Y|_{\{q=1\}})^{-1} \circ \Delta \circ (e_X|_{\{p=1\}}).$$

For $p(u) = p(v) = 1$,

$$q(\Delta_M u - \Delta_M v) = \|\Delta(e_X u) - \Delta(e_X v)\|_Y = \|e_X u - e_X v\|_X = p(u - v).$$

These are the coordinate norm models and exact sphere transport. More explicitly, for $e = e_X$ or e_Y its model record has

$$p_e(x) = \|ex\|, \quad a_e = (\max(\|e^{-1}\|_{\text{op}}, 1))^{-1}, \quad b_e = \max(\|e\|_{\text{op}}, 1).$$

Both constants are positive. The inverse bound $\|x\|_\infty \leq \|e^{-1}\|_{\text{op}} \|ex\|$ gives $a_e \|x\|_\infty \leq p_e(x)$; the direct bound gives $p_e(x) \leq b_e \|x\|_\infty$; continuity is the composition of e with the norm. These specify the seminorm, two constants, positivity, lower and upper comparisons and continuity fields. The model carrier is the same vector space with norm p_e ; the reference and model conversions are identity on values, zero and subtraction. On the sphere $p_e(u) = 1$, so $u \mapsto eu$ and $y \mapsto e^{-1}y$ give the two inverse subtype maps used in Δ_M .

3. Recover in the models and compose back. With standard Lebesgue measure on $\mathbb{R}^n$, put $K_p = \{p \leq 1\}$, $v_p = \lambda(K_p)$, and

$$\mathcal{P}_p^N = \text{conv}_{\mathbb{R}} \{\pm v_p \Pi_n(B) : B : \mathbb{R}^n \rightarrow \mathbb{R}^N, \forall x, \|Bx\|_\infty \leq p(x)\},$$

where B ranges over real linear maps and $\Pi_n(B)$ is the increasing-row maximal-minor vector. Define $\mathcal{P}_q^N$ similarly. The sphere isometry Δ_M gives equality of these bodies for every N , by sphere-induced body invariance. Input these models and all body equalities to norm-preserving linear recovery, as in sphere-metric recovery in norm models, obtaining bijective real linear L with $q(Lx) = p(x)$. The same maps give

$$A = e_Y \circ L \circ e_X^{-1}, \quad \|Ax\|_Y = q(L(e_X^{-1}x)) = p(e_X^{-1}x) = \|x\|_X.$$

The inverse is $e_X \circ L^{-1} \circ e_Y^{-1}$. This is coordinate model recovery and return. It proves existence, without claiming $A|_{S_X} = \Delta$.

4. **Restrict an ambient linear isometry for the converse.** Given A , the equations $\|Au\|_Y = \|u\|_X = 1$ and the corresponding inverse equation make its restriction a sphere bijection. For $u, v \in S_X$,

$$d_Y(Au, Av) = \|A(u - v)\|_Y = \|u - v\|_X = d_X(u, v).$$

Thus the restriction is a sphere isometry equivalence, including the empty case, as in restriction of a linear isometry.

The conclusion is an equivalence of existence statements, not an extension assertion for the supplied Δ . When both spaces are initially known to be finite-dimensional, the two-sided finite-dimensional version gives the same equivalence. The finite-dimensional-domain implication and finite-dimensional-codomain implication use the respective side of the present disjunction; the combined forward implication selects the applicable direction. These variants preserve the same existence conclusion and do not supply an agreement condition with Δ .

Main citations

- Exact declaration and source
- two-sided finite-dimensional version
- finite-dimensional-domain implication
- finite-dimensional-codomain implication
- combined forward implication
- finite dimension transfer along a sphere isometry
- equality of dimensions
- the finite-dimensional construction including zero
- the coordinate norm models
- exact sphere transport
- sphere-induced body invariance
- norm-preserving linear recovery
- sphere-metric recovery in norm models
- coordinate model recovery and return
- restriction of a linear isometry

Lean source signature (exact)

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-- If one ambient space is finite-dimensional, the unit-sphere chord metric
is a complete invariant of real normed spaces up to linear isometry. -/
theorem nonempty_isometryEquiv_iff_nonempty_linearIsometryEquiv_of_finiteDimensional
  {X : Type u} {Y : Type v}
  [NormedAddCommGroup X] [NormedSpace ℝ X]
  [NormedAddCommGroup Y] [NormedSpace ℝ Y]
  (hfin : FiniteDimensional ℝ X ∨ FiniteDimensional ℝ Y) :
  Nonempty ((Metric.sphere (0 : X) 1 ≈i Metric.sphere (0 : Y) 1)) ↔ Nonempty (X ≈i[ℝ] Y)
```

In the source	Mathematical meaning
<code>X; Y; [NormedAddCommGroup X] [NormedSpace ℝ X]; [NormedAddCommGroup Y] [NormedSpace ℝ Y]</code>	The two real normed spaces, their additive norms and real scalar actions.
<code>hfin : FiniteDimensional ℝ X ∨ FiniteDimensional ℝ Y</code>	At least one space is finite-dimensional; $\vee$ means or. Both are not required initially.
<code>Metric.sphere (0 : X) 1; Metric.sphere (0 : Y) 1</code>	S_X, S_Y , subtypes with inherited chord distances. These spheres may be empty.
<code>Metric.sphere (0 : X) 1 ≈_i Metric.sphere (0 : Y) 1</code>	A bijection $\Delta : S_X \rightarrow S_Y$ preserving $\ u - v\ _X$ and the corresponding target distance. $\approx_i$ is isometry equivalence.

In the source	Mathematical meaning
<code>X ≈_i[ℝ] Y</code>	A real linear bijection $A : X \rightarrow Y$ preserving $\ x\ _X$ as $\ Ax\ _Y$.
<code>Nonempty (...) ↔ Nonempty (...)</code>	The entire conclusion says existence of a sphere equivalence if and only if existence of an ambient linear isometry equivalence. No restriction-equals-input condition occurs.

`hfin` is a hypothesis, and $\leftrightarrow$ is the final if-and-only-if. The two occurrences of `Nonempty` quantify independently over a sphere bijection and an ambient real linear isometry. There is no equation asserting that the latter restricts to the former.

Surrounding assumptions and aliases.

▼ Exact content identity

Declaration: `MathlibAnnex.Sphere.nonempty_isometryEquiv_iff_nonempty_linearIsometryEquiv_of_finiteDimensional`

Accepted content SHA-256: `ab2edc84df9478a9f7988f6b95aa42ee5a995655cf46a24d6111a767b341ec1c`

Accepted source guide SHA-256: `ea58aee79f80e280d7b7720a69bb8b5f3bd83599e3197f1490f0288074bc581d`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`